

코딩 테스트 대비 알고리즘 스터디

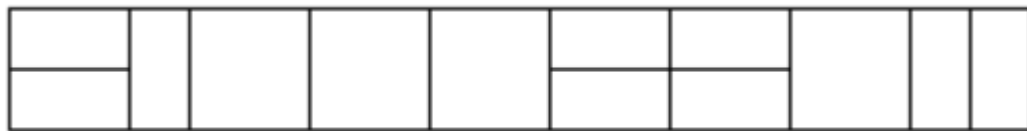
5회차: DP 알고리즘 (2)

11727: 2 x n 타일링 2

문제

2×n 직사각형을 1×2, 2×1과 2×2 타일로 채우는 방법의 수를 구하는 프로그램을 작성하시오.

아래 그림은 2×17 직사각형을 채운 한가지 예이다.



2x1

1x2

2x2

입력

첫째 줄에 n 이 주어진다. ($1 \leq n \leq 1,000$)

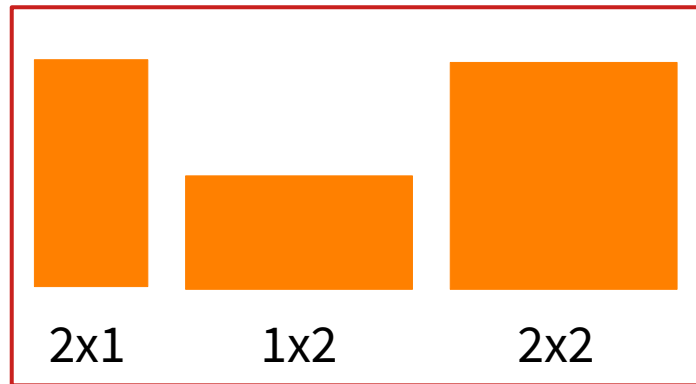
출력

첫째 줄에 2×n 크기의 직사각형을 채우는 방법의 수를 10,007로 나눈 나머지를 출력한다.

문제 접근법

- 전형적인 점화식 문제

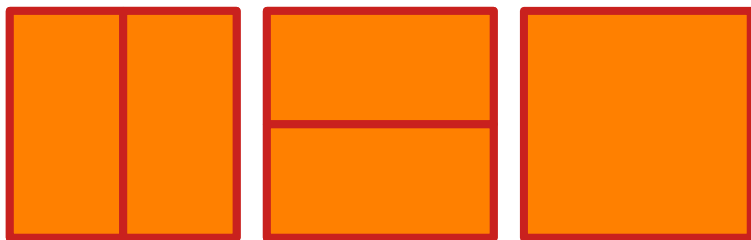
- 가장 간단한 경우부터 나열해 보면서 패턴 분석



$N = 1$



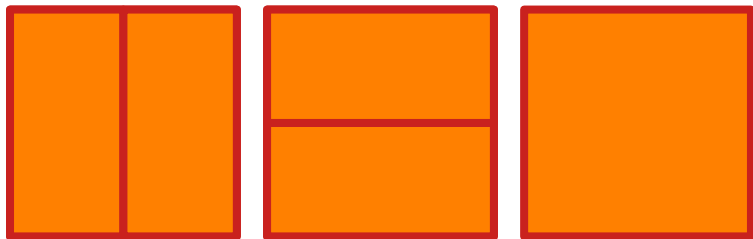
$N = 2$



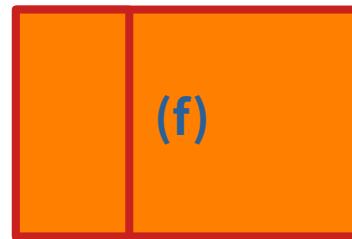
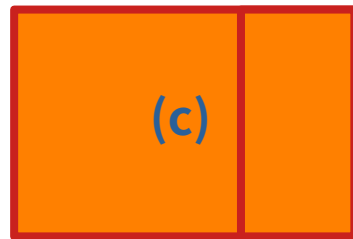
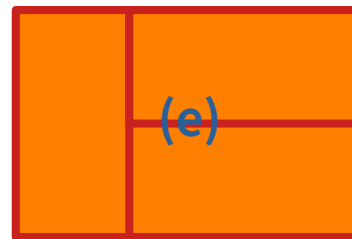
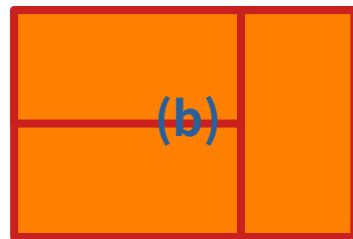
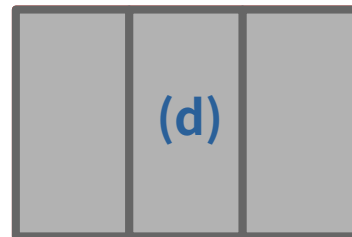
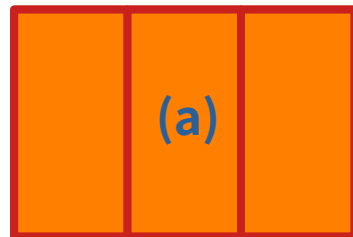
$N = 1$



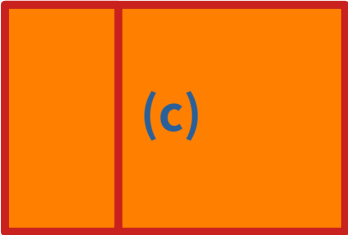
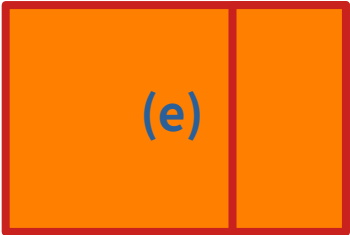
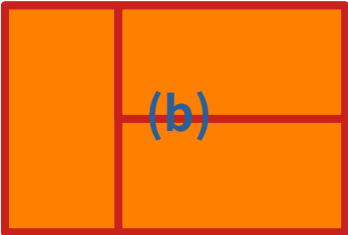
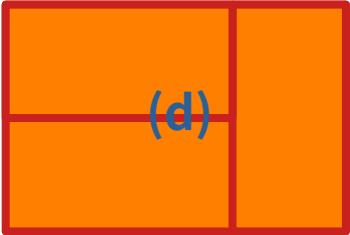
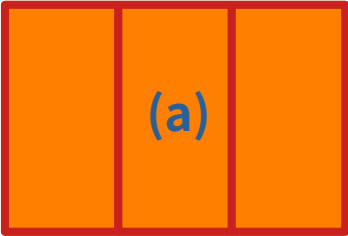
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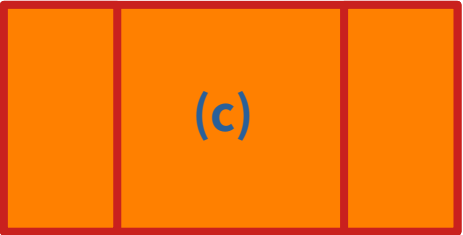
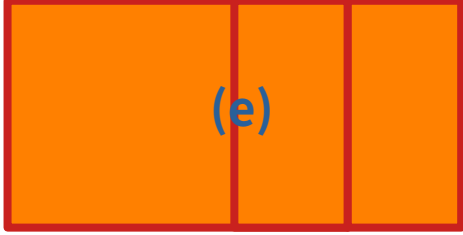
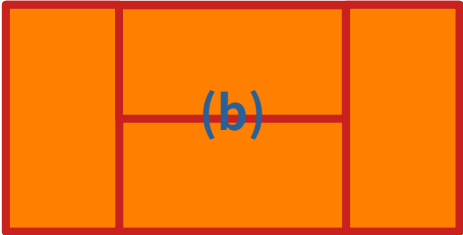
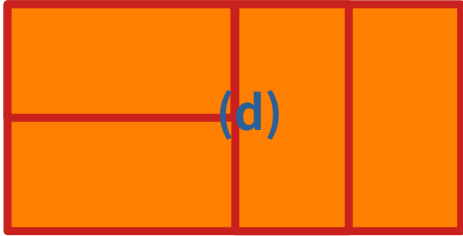
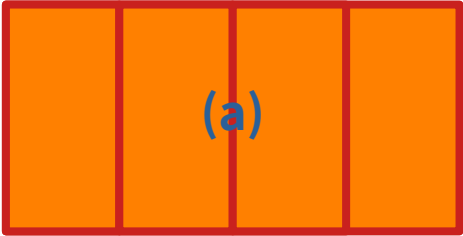
$N = 3$



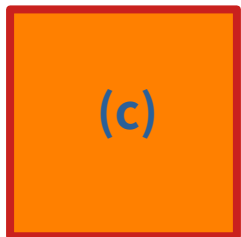
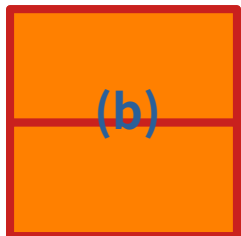
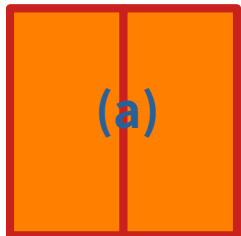
N = 3



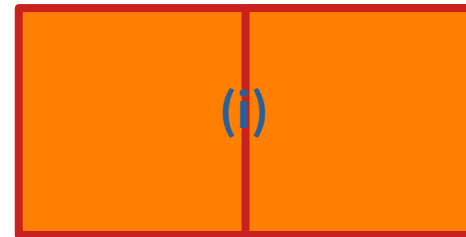
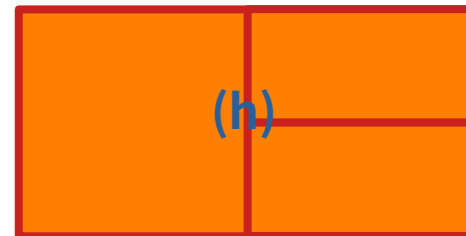
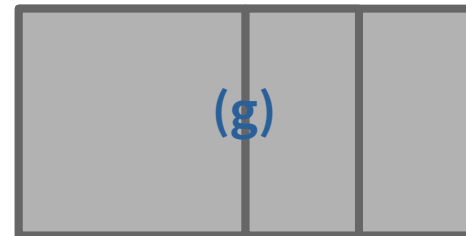
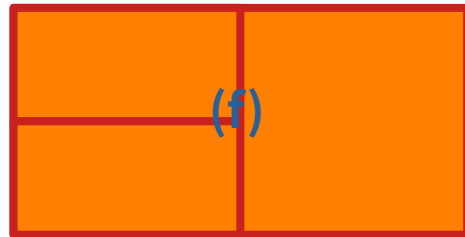
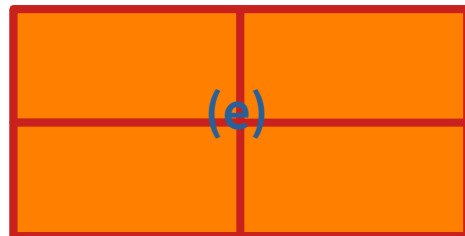
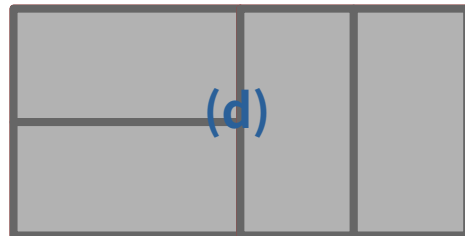
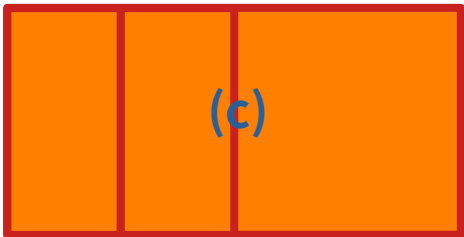
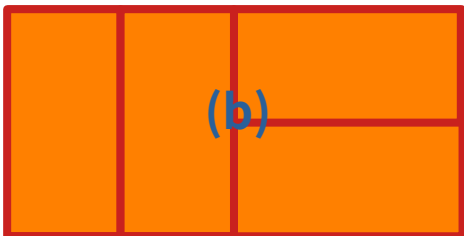
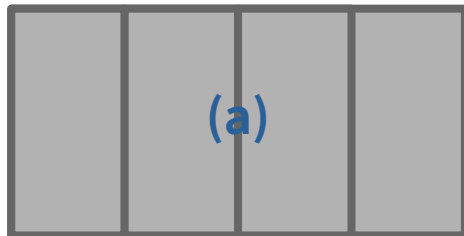
N = 4



N = 2



N = 4

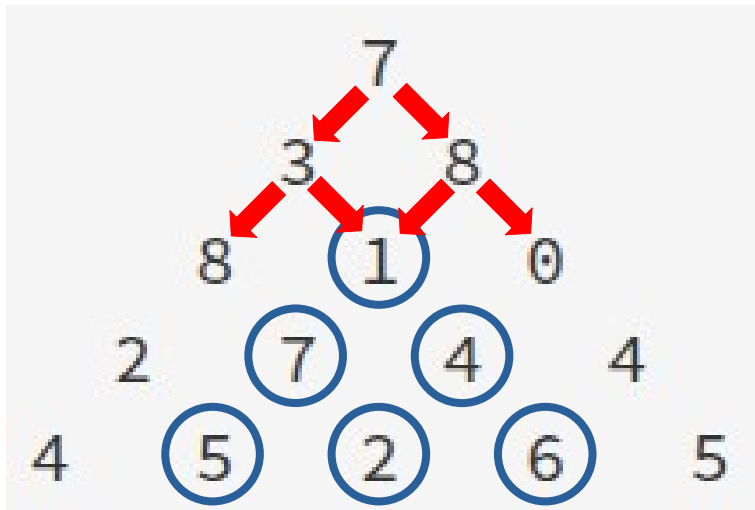


dp(N-1)

점화식: $dp(N) = dp(N-1) * dp(1) + dp(N-2) * 2 = dp(N-1) + 2 * dp(N-2)$

1932: 정수 삼각형

문제

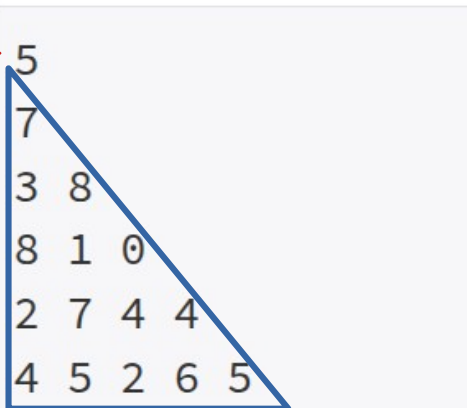


출력

첫째 줄에 합이 최대가 되는 경로에 있는 수의 합을 출력한다.

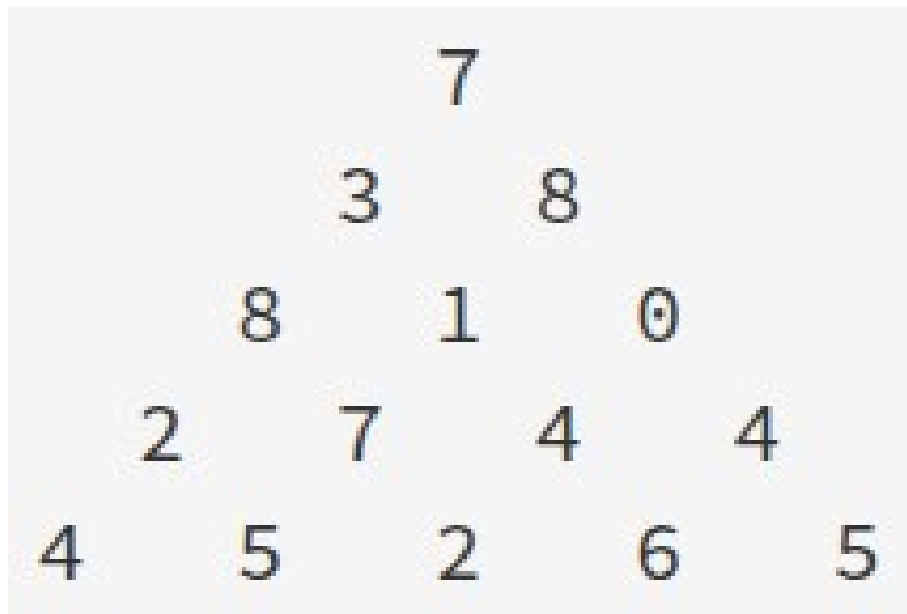
예제 입력 1 복사

삼각형의 크기 →



DFS 또는 BFS 접근법:

모든 경로를 “전부” 탐색한 뒤, 최적해
도달 (Brute-Force)



7				
3	8			
8	1	0		
2	7	4	4	
4	5	2	6	5

7				
7+3 = 10	7+8 = 15			
10+8 = 18	10+1, 15+1 = 16	15+0 = 15		
18+2 =20	18+7, 16+7 = 25	16+4, 15+4 = 20	15+4 =19	
20+4 =24	20+5, 25+5 = 30	25+2, 20+2 = 27	20+6, 19+6 = 26	19+5 = 24